

Requirements Document

HALO: High Autonomous Low-SWaP Operations

Team Members

Sloan Hatter (shatter2022@my.fit.edu)

Blake Gisclair (bgisclair2022@my.fit.edu)

Faculty Advisor

Dr. Ryan T. White (rwhite@fit.edu)

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1. Introduction

1.1. Purpose

The purpose of this document is to outline the functional and non-functional requirements of HALO, a neural network aimed to be deployed on small computers to carry out orbital vision tasks. The requirements herein include real-time object detection, low SWaP deployability, binary quantization, and performance metrics; functional and non-functional, respectively.

1.2. Scope

The scope of this project is to achieve a 1-bit representation of neural network weights through binary quantization with the goal of efficiently running a neural network on a small-scale computer that can be utilized on satellite systems. HALO shall be able to efficiently carry out orbital object classification, segmentation, and detection while running on a small-scale computer, such as a Raspberry Pi or Jetson.

1.3. Definitions

HALO: High-Efficiency Low-SWaP Operations

ViT: Vision Transformer

A machine learning model in which the transformer architecture, typically used for processing sequential data for language and text tasks, is adapted for computer vision tasks.

Low-SWaP: Low Size, Weight, and Power

A computer that is designed to be small, lightweight, and energy-efficient. These types of computers are typically used in drone technology, military and aerospace applications, and embedded systems. Examples of these devices include Raspberry Pis and Jetsons.

Binary Quantization

A neural network compression technique that reduces each value of a model's weights and activations down to a single bit, either 0 or 1.

Classification, Segmentation, and Detection in Neural Networks

Classification: assigns a single category label to an image.

Segmentation: assigns a class label to each pixel within an image.

Detection: identifies specific objects within a given image space.

1.4. References

This document used the “IEEE Recommended Practice for Software Requirements Specifications” document as a framework reference.

2. Overall Description

2.1. Project Perspective

HALO is a neural network and low-SWaP computer duo that performs vision tasks, namely classification, segmentation, and detection, on orbital objects. This model is intended to be deployed on on-orbit satellites for use in autonomous systems. HALO aims to make orbital object vision tasks more efficient and less resource-intensive by running on low-SWaP computers, such as a Raspberry Pi or Jetson. Running HALO on a smaller computer will reduce the need to deploy large computers on satellite systems, reducing the amount of space and power normally required. The HALO neural network will also be reduced in order to run efficiently on a smaller computer. Reduction will be done through binary quantization, a method that heavily reduces the storage size of vectors, improves memory efficiency, and speeds up search performance.

2.2. Product Functions

Real-Time Data Processing

HALO shall be able to process real-time data received via an on-board camera feed. For real-world simulation purposes, this camera feed will be a video taken of a digital twin satellite.

Object Detection Capabilities

HALO shall be able to accurately detect objects, more specifically orbital satellite objects, within a given image space. The system shall be able to draw bounding boxes around detected objects and classify them accordingly.

Low-SWaP Deployability

HALO shall be able to be deployed on a low-SWaP computer, ideally a Raspberry Pi, but if needed, a Jetson. This is a main feature of the system since the goal is to limit the amount of space and resources orbital object detection systems currently take up by running orbital object detection on a smaller computer that takes up less space and resources.

2.3. User Characteristics

This project is targeted towards Dr. Ryan T. White for future use in work with the United States Air Force.

2.4. Constraints

This project will be run on GPUs provided by the Florida Tech NETs lab on campus. Many of these GPUs are being used for various other projects, so

runtime slots will have to be worked out to ensure the project obtains the resources it requires.

3. Functional Requirements

3.1. Model Inference

In order for HALO to be able to process data in real-time, the model's frame rate needs to be increased. A higher frame rate corresponds to lower latency per frame, which brings the model's performance closer to real-time operation. This improvement can be achieved by reducing the model's inference time, which is the time it takes for a model to process a single frame. Reducing the model's inference can be done through quantization. Quantization decreases a model's size, which in turn decreases the amount of data the model needs to process, leading to faster computations and ultimately an increased frame rate. The increase in frame rate will allow the model to achieve real-time data processing.

3.2. Vision Transformer Utilization

In order for HALO to be able to perform any sort of vision task, including classification, segmentation, or detection, the model will have to utilize some form of neural network visualization technique or tool. The visualization tool selected for this project is the vision transformer neural network architecture. This architecture will be utilized by our neural network model to perform the aforementioned vision tasks. Utilizing a ViT compared to a Convolutional Neural Network (CNN) is much more appropriate for this project, because, unlike CNNs, ViTs are able to capture global context and relationships within images, leading to increased accuracy and robustness when used on large datasets. ViTs are also more flexible and scalable, as their architecture is adaptable to a wider variety of vision tasks and is able to handle more complex problem sets. Thus, utilizing the ViT framework is ideal for this project, as it will give HALO the ability to perform at its optimal capabilities.

3.3. Binary Quantization

In order for HALO to run efficiently on a low-SWaP computer device, the model will need to be reduced. The reduction method chosen for HALO is model quantization, more specifically, binary quantization. Model quantization is a reduction technique carried out on neural networks to reduce the model's size and resource consumption by decreasing the model's weight representations. Subsequently, this technique also serves as an optimizer, as reducing a model's bit representation speeds up inference and lowers resource consumption. The aim of this project is to use a specific type of model quantization, known as binary quantization. Binary quantization converts a model's weights and activations into a binary value, either 0 or 1, which in turn drastically reduces the model's memory footprint, speeds up inference, and lowers power and resource consumption. This quantization method is required to be able to efficiently deploy HALO onto low-SWaP hardware.

4. Non-Functional Requirements

4.1. Performance Requirements

Performance metrics include power consumption, frame rate, and model accuracy. Listed below are the ideal performance requirements of HALO:

Wattage: The ideal wattage pull of HALO is between 10-20, with 15 being ideal.

Frame Rate: The ideal frame rate of HALO is 30 FPS.

Accuracy: The ideal accuracy of HALO is approximately 0.4.